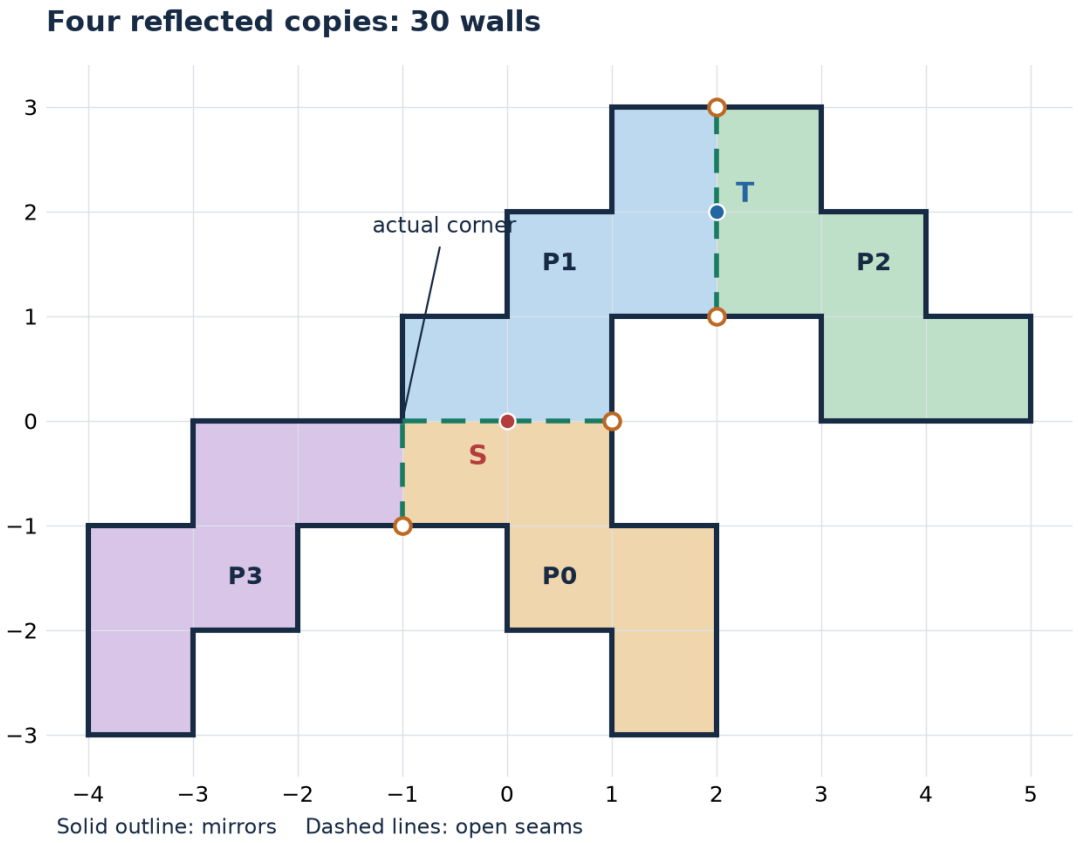
A second room: 30 walls

Four reflected copies of the same five-square base, with S = (0,0) and T = (2,2).



Add one reflected copy on the left

Keep P0, P1 and P2 from the 22-wall construction. Add P3 = {(-2-x,y) : (x,y) in P}. This reflects P0 across x = -1. Its interior is disjoint from the other three copies. It attaches to P0 on the interval {-1} x [-1,0].

The new union is a strictly simple orthogonal polygon with 30 maximal walls: 17 convex corners and 13 concave corners. S and T are still interior points. The new open seam is green and dashed; its lower endpoint (-1,-1) is a smooth wall point.

At (-1,0), three copies meet in three quadrants. The missing fourth quadrant makes this an actual 270-degree corner, so a valid billiard path cannot hit it. There is no pinch or hole. This is a different four-copy layout with 30 walls.

Fold P3 back by (x,y) -> (-2-x,y), together with the previous three folding maps. All the maps agree at every shared point. The same base obstruction will prove that S and T remain dark.

Why the second room is dark

The base P is unchanged: its only noncorner integer points are the even-even marks a and b. Unfolding any ordinary a-to-b path produces an even-even nonzero endpoint. Repeated halving forces an odd-coordinate integer point earlier on that straight segment. Folding it back gives an actual base corner. Thus no such ordinary base path exists.

The four smooth seam-endpoint events

Smooth point in the 30-wall room	Convex corner in P
(-1,-1)	(-1,-1)
(1,0)	(1,0)
(2,1)	(2,-1)
(2,3)	(2,-3)

At each listed point, a physical wall reflection switches the adjacent copy. Folding therefore reverses both velocity components, exactly as in the 22-wall proof. Every other preimage of a base corner is an actual room corner. No concave base corner becomes a smooth or interior room point.

For a precise count, the 40 base-corner incidences consist of 17 convex room corners, 12 ordinary concave room corners, three incidences at the actual concave corner (-1,0), and eight incidences at the four smooth points in the table: 17 + 12 + 3 + 8 = 40.

Now suppose an S-to-T path existed. Fold it, stop at its first b and start at its last a before that b. A first corner reversal would retrace to a before reaching b, contradicting that choice. With no reversal, the resulting a-to-b path would be ordinary, contradicting the lattice argument. Hence S and T do not illuminate each other.

All preimages of the two marks

a = (0,0)	(-2,0), (0,0), (4,0)
b = (2,-2)	(-4,-2), (2,-2), (2,2)

The same first-b/last-a choice handles these extra smooth-boundary preimages. The finite-event argument is unchanged because all seams are horizontal or vertical and a connecting path would be nonaxial.

Cyclic coordinates of the 30-wall boundary

(-4,-3), (-4,-1), (-3,-1), (-3,0), (-1,0), (-1,1)
(0,1), (0,2), (1,2), (1,3), (3,3), (3,2)
(4,2), (4,1), (5,1), (5,0), (3,0), (3,1)
(1,1), (1,-1), (2,-1), (2,-3), (1,-3), (1,-2)
(0,-2), (0,-1), (-2,-1), (-2,-2), (-3,-2), (-3,-3)

Thus the four-copy construction gives a second simple mirror room with a dark pair of interior points.